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Reaction to the first number of FLEA NEWS has been most favourable, which is encouraging. Thanks are due to those who took the trouble to write.

Please find herewith a list of additional 1973 literature, the first 1974 titles and a list of the recipients of FLEA NEWS which will serve as a mailing list of flea specialists and of others who are interested in some aspect of the study of fleas.

F.S.

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FIRST EUROPEAN CONFERENCE ON FLEAS (preliminary notification)

Most students of fleas are in agreement about the potential value of a general meeting to discuss various aspects of our subject of study. It is gratifying, therefore, to announce that The Zoological Institute of the University, Lund, Sweden -at the instigation of its director, Professor Per Brinck- will be organizing the first such conference to be held in Europe. This conference, supported by a grant from the Swedish Natural Science Research Council, will last three or four days during the first half of June, 1975; it is intended to be of a rather informal nature and will take place in the Zoological Institute, Lund, and/or at its Stensoffa Field Station (near Lund).

The Zoological Institute of the University, Lund, is honoured and privileged to invite anyone who is seriously interested in the study of fleas to attend the conference. As the venue is in Europe, it should be realized that European/Palaearctic themes (apart from general ones) may dominate lectures and discussions.

Those who have access to funds will no doubt apply for a grant to cover the cost of travelling and boarding. Grants to cover the basic expenses connected with the stay in Lund will be available for participants from eastern Europe.

Accommodation: by choice either at the Stensoffa Field Station (about 20 beds) in hostel conditions @ 10 Sw.Kr. a night (add for meals at least 25 Sw.Kr. a day) or at an hotel in Lund (more expensive, more comfort).

The organizing committee consists of Mrs G. Brinck-Lindroth, Mr L. Lundqvist (both of The Zoological Institute, Lund) and Mr F.G.A.M.Smit (British Museum (Natural History)).

If you would like to attend the conference, please inform (without much delay) Mrs G. Brinck-Lindroth, Department of Animal Ecology, University of Lund, Ecology Building, S 223 62 Lund, Sweden; applicants from eastern Europe, please state at the same time whether or not you require a subsistence grant.

Details about the meeting will be circulated in due time.

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What would be a neat, single, universal name for "the study of fleas" and for a person who is 'doing fleas'? At present "fleaology" is of course only jocularly used in English [in analogy with the correct German "Flohkunde"] while the researcher is referred to as a "student of fleas" (a phrase often used by the late Dr Karl Jordan). "Siphonapterology" and "siphonapterologist" (or the incorrect shortened version "siphonapterist") have been used, but those names are rather mouthfuls. ["Siphonaptera" as the name for the Order is at long last in universal use; the very few authors still occasionally referring to "Aphaniptera" are behind times].

What about a name based on either the Greek or Latin word for 'flea', i.e. "psullos" latinized as "psylla", or "pulex"? - "Psyllology" and "psyllologist" are not ideal because the stem of the word is similar to that of the Hemipteran Psyllidae and because the peculiarity of spoken English not to pronounce p before s at the beginning of a word [an "unscholarly practice" (Oxf. Dict.)] makes those words sound rather psilly. - "Pulicology" and "pulicologist" (although bastard words) would seem quite usable.

Any views or suggestions, please?

F.S.

How to draw setae (and smoothly curved lines) - For the benefit of the rather surprising number of colleagues who do not know:

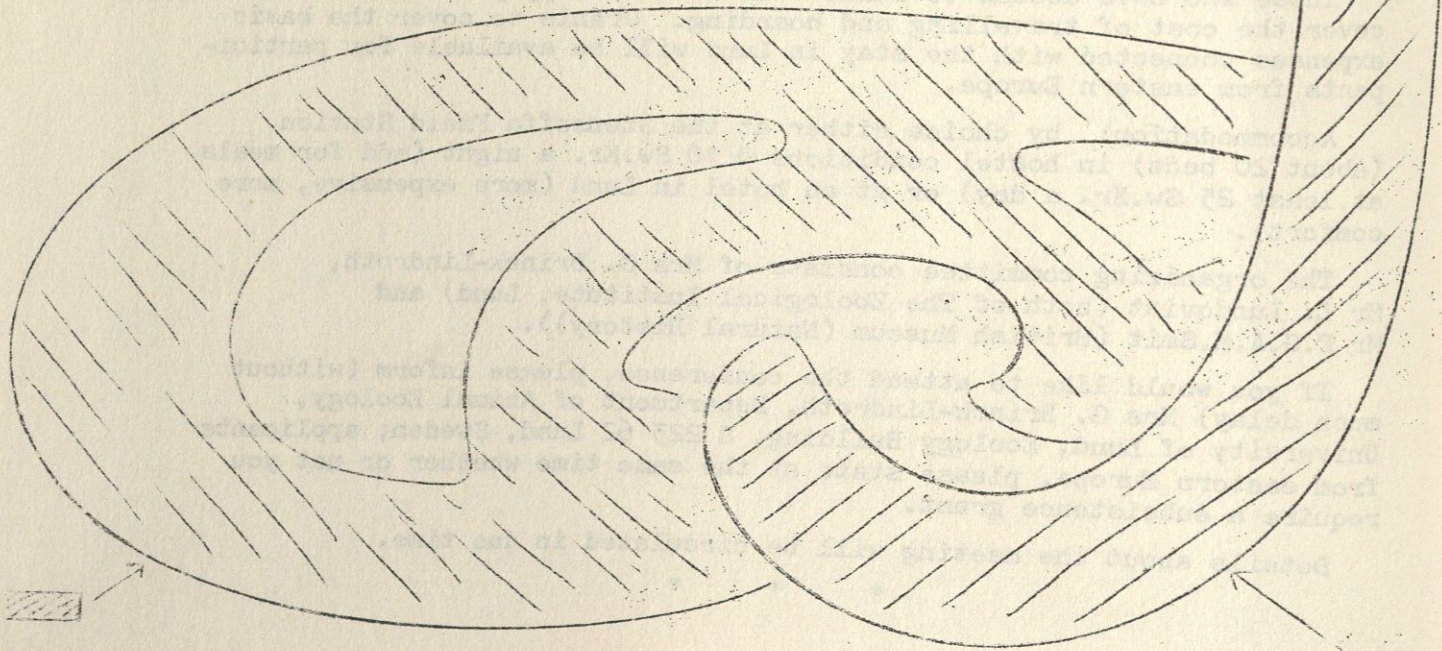
Setae (being usually at least very slightly curved) can easiest be drawn with the aid of "French curves" or the more recently introduced flexible rulers. French curves should either have a bevelled edge (as the old ivory J-shaped one which Dr Jordan used for many years; see figure on the right) or be at least a little over 2 mm thick (like the perspex one which I have been using for over 25 years; see figure below). Unfortunately most French curves now on sale are too thin for use with a nib-pen. Rapidograph (tube) pens can perhaps be used with the thin plastic curves, but no draughtsman worth his salt will ink flea drawings with a Rapidograph (they are fine for e.g. writing slide labels). - Small setae can be drawn easily free-hand.

About free-hand drawing: the nib of the pen should be at a right angle with the line being produced by moving the nib backwards; the entire lower arm and hand (with hardly any movement in the wrist) should rest on the surface of table or desk - the elbow is the pivot. Do not -or hardly- shift the position of the arm/hand but instead shift the drawing to follow the pencil lines.

F.S.

"...you must not work with haste - rather with great enjoyment and pleasure. And I give you this advice, that the day before the day you want to work at this job [delicate drawing], you hold your hand to your neck, or in your bosom, so as to get it all unburdened of blood and weariness."

Cennini, 1450, Il Libro dell'Arte



LITERATURE ON SIPHONAPTERA PUBLISHED IN 1973 (list 2)

- Bahmanyar, M. - Carnivores as sentinels in plague surveys. Abstr. 9th int. Congr. trop. Med. Mal. (Athens) 1: 41-42.
- Balashov, Yu.S. & Daiter, A.B. - Bloodsucking arthropods and Rickettsiae. Leningrad. 250 pp., 199 figs. [Siph.: 170-191, figs. 167-192]
- Beaucournu, J.C. - Quatre siphonaptères nouveaux pour la faune française. Description de *Ctenophthalmus* (C.) andorrensis catalaniensis ssp.nov. Bull. Soc. scient. Bretagne 47(3-4): 169-175, figs. 1-5.
- Beaucournu, J.C. - Notes sur les siphonaptères parasites de carnivores en France. Ann. Parasit. hum. comp. 48(3): 497-516, fig. 1.
- Becker, K. - Zur Kenntnis der Flöhe Deutschlands. [review]. Prakt. SchädliBekämpf. 25(3): 26-27
- Beesley, W.N. - Control of arthropods of medical and veterinary importance. Adv. Parasit. (Lond.) 11: 115-192, figs. 1-3.
- Bruce-Chwatt, L.J. - Global problems of imported disease. Adv. Parasit. (Lond.) 11: 75-114, figs. 1-4.
- Buchalski, L. - The usefulness of phosphororganic compound IPO-63 in the control of some ectoparasites of domestic animals. Medycyna wet. (Lublin) 29(10): 602-604.
- Campos, S.L. & Peña, G.L.E. - Los insectos de isla de Pascua. Rev. chil. Ent. 7:216-229.
- Cavanaugh, D.C., Hetrick, M., Harrison, D.N. & Marshall, J.D. - Plague survey and surveillance: a critical evaluation of current procedure recommended to detect plague in rodent populations. Abstr. 9th int. Congr. trop. Med. Mal. (Athens) 1: 40.
- Chow, C.Y. - Arthropods of public health importance in Korea. Korean J. Ent. 3(1):31-54.
- Denis, J.R. & Bitsch, J. - Structure céphalique dans les ordres des insectes. Ordre des Aphaniptères. in: P.P.Grassé - Traité de Zoologie 8(1): 547-554, figs. 304-307.
- Fain, A. & Beaucournu, J.C. - Deux nouveaux hypopes vivant en association phorétique sur des puces de carnivores (Acarina: Sarcoptiformes). Acarologia 15(1): 138-143, figs. 1-7.
- Fain, A. & Beaucournu, J.C. - Description de trois nouveaux hypopes d'Anoetidae phorétiques sur des puces de mammifères. Acarologia 15(3): 514-520, figs. 1-6.
- Greenwood, M.T. - An introduction to the collection of fleas of small mammals. J. biol. Educ. 7(6): 5-14, figs. 1-10.
- Haas, G.E. - Morphological notes on some Siphonaptera (Leptopsyllidae and Ceratophyllidae of New Mexico. Amer. Midl. Nat. 90(1): 246-252, figs. 1-11.
- Haas, G.E. & Walton, D.W. - Fleas (Siphonaptera) infesting small mammals from the western Oriental Region. Korean J. Parasit. 11(2): 102-107.
- Haas, G.E. & Wilson, N. - Siphonaptera of Wisconsin. Proc.ent.Soc.Wash. 73(3):302-314.
- Karimi, Y. - Recherches sur la peste au Brésil - recherche écologique dans le Nordeste. Abstr. 9th int. Congr. trop. Med. Mal. (Athens) 1: 42.
- Lapierre, J. & Holler, C. - Notions épidémiologiques et cliniques de parasitologie. Concours méd., Paris 44: 48, figs. 1-16.
- MacDonald, M.A. - A new station for the flea *Ornithopsylla laetitiae* Rothschild in Scotland. Entomologist 106(1323): 169.
- Méndez, E. & Haas, G.E. - *Megarhroglossus wilsoni*, new species, with notes on the genus in New Mexico (Siphonaptera: Hystrihopsyllidae). Ann. ent. Soc. Amer. 66(5): 1129-1139, figs. 1-37.
- Mollaret, H.H. - Etat actuel des connaissances sur les causes de l'invétération de la peste dans les foyers naturels. Abstr. 9th int. Congr. trop. Med. Mal. (Athens) 1: 40-41.

(1973)

- Ono, Z. - Ectoparasites from small mammals on the Nopporo Forest Park near Sapporo, Hokkaido, Japan. Rep. Hokkaido Inst. publ. Hlth (23): 49-56. [in Japanese]
- Ono, Z. & Takada, N. - Siphonaptera from small mammals in northern Honshu, Japan. Res. Bull. Meguro parasit. Mus. (7): 28-31.
- Peenen, P.F.D. van et al. - Mammals and their parasites in Gumbasa valley, central Sulawesi, Indonesia. Abstr. 9th int. Congr. trop. Med. Mal. (Athens) 2: 95.
- Řeháček, J. - Organ, tissue and cell cultures of haematophagous arthropods and their application in virology and rickettsiology. Ent. Probl., Bratislava 11: 9-55, figs. 1-28.
- Rodriguez, R. - Infestacion por *Dipylidium caninum* a la especie humane. Rev/ta ibér. Parasit. 33(2-3): 449.
- Rothschild, M., Schlein, Y., Parker, K., Neville, C. & Sternberg, S. - The flying legs of the flea. Scient. Am. 229(5): 92-98, 100, figs.
- Shumilo, R.P. & Lunkashchu, M.I. - Fleas (Aphaniptera) of terrestrial birds of Moldavia and the Chernovtsy oblast. Parazity Zhiv. Rast. (Kishinev) 9: 208-214.
- Steele, J.H., Arambulo III, P.V. & Beran, G.W. - The epidemiology of zoonoses in the Philippines. Archs envir. Hlth 26(6): 330-339.
- Suciu, M. - Catalogue of the Siphonaptera from Rumania. Studii Comun. Muz. St. Nat. Bacau 6: 47-72, 1 map.
- Suciu, M. - *Ctenophthalmus agyrtes romanicus* n.ssp. (Siphonaptera: Hystrichopsyllidae) from Romania. Rev. roum. Biol., Zool. 18(3): 181-184, figs. 1-2.
- Suciu, M. - Quelques considérations zoogéographiques sur les puces (Siphonaptera) de Roumanie en fonction de leurs hôtes. Rev. roum. Biol., Zool. 18(2): 111-118.
- Suciu, M. - *Ctenophthalmus agyrtes* Heller in România. Studii Cerc. Biol., Zool. 25(4): 311-321, figs. 1-10.
- Tiik, L. - The plague on Hiiumaa island in 1910-11. Eesti Loodus 10: 617-620.
- Velimirovic, B. - Reappearance of plague in Khmer Republic (formerly Cambodia). Z. Tropenmed. Parasit. 24(3): 265-270, figs. 1-2.
- Velimirovic, B. - Surveillance of small mammals and flea vectors in a plague-infected port. Z. Tropenmed. Parasit. 24(4): 539-548, figs. 1-3.
- Velimirovic, B. - Review of the epidemiological situation of plague since the last Congress on Tropical Medicine and Malaria in 1968. Abstr. 9th int. Congr. trop. Med. Mal. (Athens) 1: 39.
- Walker, M.L., Rayburn, J.D., Humphrey, J.M. & Segal, D.B. - Parasite-subject catalogue subject headings and treatment. U. S. Dept Agric. Index-Cat. med. vet. Zool. 18(6): 1-339.
- Whitaker, J.O. - External parasites of bats in Indiana. J. Parasit. 59(6): 1148-1150.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1974 (list 1)

- Askew, R.R. - Insects from Bardsey Island. Ent. Gaz. 25(1): 45-51. (Siph.: 50).
- Baker, J.R. - Protozoan parasites of the blood of British wild birds and mammals. J. Zool., Lond. 172(2): 169-190, figs. 1-7.
- Beaucournu, J.C. - Technique de récolte des Siphonaptères. Entomologiste 30(2): 80-86.
- Beaucournu, J.C. - Sur les Typhloceras et Atyphloceras de France (Siphonaptera, Hystrichopsyllidae). Ann. Soc. ent. Fr. (N.S.) 10(1): 129-148, figs. 1-76.
- Bibikova, V.A. & Klassovsky, L.N. - Transmission of plague by fleas. (Meditcina). 190 pp.
- Christov, L. - Aphaniptera from insectivores and rodents in the Rozovo valley [Bulgaria]. Izv. zool. Inst. Muz., Sofia 39: 195-205.
- Clark, P.H. & Cole, M.M. - Oriental rat fleas: evaluation of three systemic insecticides in baits for control on cotton rats in outdoor pens. J. econ. Ent. 67(2): 235-236.
- Fuller, G.K. - Observations on flea attachment at low hair densities on man. J. nat. Hist. 8(2): 207-213, figs. 1-2.
- Gaaboub, I.A. & Abu-Hashish, T.A. - Seasonal abundance of fleas and their rat hosts in Alexandria region, Egypt. Z. angew. Ent. 75(2): 204-212, figs. 1-5.
- Gershkovich, N.L. & Naumov, R.L. - Fleas (Siphonaptera) of small mammals and passerine birds in a focus of tick-borne encephalitis in western Sayan. Byull. mosk. Obshch. Ispyt. Prir. (Biol.) 79(2): 39-49.
- Kir'yakova, A.N., Amirov, Zh.A., Byesedin, B.D., Pasternak, Ye.P., Kamyshev, A.I., Loseva, Ye.I. & Byekmambetova, A.V. - An evaluation of the effectiveness of field disinsectisation in the Cis-Aral Karakumy. Parazitologiya 8(2): 157-163.
- Korneyev, V.A., Gibet, L.A. & Nikiforov, L.P. - The ecology of *Leptopsylla silvatica* (Aphaniptera) in the Mariisky A.S.S.R. Zool. Zh. 53(1): 66-76.
- Kozlov, M.P. & Rozanova, G.N. - The dependence of block-formation in fleas on certain aspects of the metabolism of carbohydrates in the plague microbe. Parazitologiya 8(3): 254-260.
- Li, K.-c., Chen, N.-y. & Wei, S.-f. - On two new species of Macrostylophora (Siphonaptera: Ceratophyllidae). Acta ent. sinica 17(1): 112-116, figs. 1-6.
- Liu, C.-y., Tsai, L.-y. & Wu, W.-c. - A new genus and species of Siphonaptera from Chinghai province, west China. Acta ent. sinica 17(1): 102-111, figs. 1-11.
- Miller, D.H. & Benton, A.H. - An annotated list of the Siphonaptera of Connecticut. Jl. N.Y. ent. Soc. 81(4): 210-213.
- Ovedzykhovskiy, G.A. & Barayeva, G.M. - Distribution and numerosity of fleas of *Citellus pygmaeus* in the north-east of the Volga-Ural sandy regions. Zool. Zh. 53(4): 585-593.
- Reitblat, A.G. & Byelokopytova, A.M. - On cannibalism and predation in flea larvae. Zool. Zh. 53(1): 135-137.
- Syba, J. & Šebek, Z. - *Leptopsylla taschenbergi calamana* Jordan, 1951 and some other findings of fleas (Insecta, Siphonaptera) in Switzerland. Folia parasit. 21(1): 93-94.
- Witt, F.G.A.M. - Notes on two fictitious fleas from Nepal. Ent. NachrBl., Wien 19(3): 130.
- Witt, F.G.A.M. - The flea *Palaeopsylla soricis* as intermediate host for the cestode *Hymenolepis scutigera*. Ent. Ber., Amst. 34(5): 94-95, fig. 1.
- Witt, F.G.A.M. & Rosický, B. - Siphonaptera from eastern Afghanistan. Folia parasit. 21(1): 29-37, figs. 1-7.
- Yang, D.-c. - A new species of the genus *Neopsylla* Wagner (Siphonaptera: Hystrichopsyllidae). Acta ent. sinica 17(1): 117-119, figs. 1-3.

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It will be understood that -although their titles are given in English above- papers in Russian or Chinese periodicals are in Russian or Chinese.)

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BULGARIA - L.Christov

CANADA - G.P.Holland

CANAL ZONE PANAMA - E.Méndez

CHINA - K.-c.Li, C.-y.Liu

CZECHOSLOVAKIA - K.Hůrka, M.Jurík,
B.Rosický, J.Ryba

EGYPT - H.Hoogstraal

ENGLAND - M.J.Cotton, R.S.George,
D.A.Humphries, M.Rothschild,
F.G.A.M.Smit

FINLAND - J.Viramo

FRANCE - J.C.Beaucournu

GERMANY - F.Peus, H.Weidner

HUNGARY - I.Szabó

JAPAN - Z.Ono

JUGOSLAVIA - S.Brelih

MEXICO - A.Barrera

NETHERLANDS - E.van den Broek

NEW ZEALAND - R.L.C.Pilgrim

NORWAY - R.Mehl

POLAND - K.Bartkowska, R.Haitlinger,
J.Lachmajer, W.Skuratowicz

PORTUGAL - H.Ribeiro

ROMANIA - M.Suciu

SCOTLAND - G.M.Dunnet, D.K.Mardon

SENEGAL - J.M.Klein

SWEDEN - P.Brinck, G.Brinck-Lindroth

SWITZERLAND - V.Aellen, W.Büttiker,
V.Mahnert

U. S. A. - K.Baker, A.H.Benton, R.E.Elbel,
A.Farhang-Azad, G.E.Haas,
M.W.Hastriter, C.E.Hopla,
E.W.Jameson, L.Kartman,
R.E.Lewis, B.C.Nelson,
F.J.Radovsky, H.E.Stark,
V.J.Tipton, R.Traub, N.Wilson

U.S.S.R. - A.N.Alekseyev, G.A.Avetisyan,
V.A.Bibikova, N.F.Darskaya,
A.I.Goncharov, E.V.Isayeva,
R.B.Kosminsky, N.T.Kunitskaya,
B.A.Rostigayev, O.I.Scalon,
Ye.F.Sosnina, V.Ye.Tiflov,
N.A.Violovich, S.A.Vysotskaya,
N.D.Yemel'yanova, V.I.Yurkina,
I.F.Zhovtyi

VENEZUELA - C.E.Machado-Allison

WEST INDIES - I.Fox

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25.V.1974